

Safaet Jaman Arman

AI Engineer ■ ML Researcher ■ NLP & Generative AI

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🌐 LinkedIn | 🐙 GitHub | 🏆 Kaggle | 📁 Portfolio

PROFESSIONAL SUMMARY

AI Engineer with production experience shipping **Retrieval-Augmented Generation (RAG)** systems for multilingual conversational products, alongside hands-on work in Generative AI, LangChain-based LLM pipelines, and TensorFlow-powered machine learning models. Led the design of a two-stage retrieval pipeline (bi-encoder + cross-encoder reranking) with a data-calibrated refusal gate while directing a 4-person delivery team. **Published researcher at ECCT 2026 (Taylor & Francis)** — developed the R-Blend ensemble framework for Antimicrobial Resistance prediction, achieving **F1-score of 0.9421 and ROC-AUC of 0.9656 across 12 clinical datasets**. Skilled in data preprocessing, NLP, model evaluation, and cross-functional collaboration in Agile environments. Passionate about applying AI to real-world problems and growing in a learning-focused, innovation-driven team.

WORK EXPERIENCE

Technical Lead & AI Engineer | Neuralis AI (Bengali & Multilingual RAG Chatbots)

Aug 2026 – Present

Part-time (~25 hrs/wk)

- Led a **4-person cross-functional team** (frontend, 2× data-collection, deployment) end-to-end while owning the ML/backend implementation directly.
- Architected a **two-stage retrieval pipeline** (bge-m3 bi-encoder + bge-reranker-v2-m3 cross-encoder) with LLM-based query rewriting and custom chunking over **bge-m3** multilingual embeddings (1024-dim).
- Implemented a **data-calibrated refusal gate** with score-gated, context-only grounded generation and inline citations to suppress hallucinated answers on out-of-scope queries.
- Designed a regex-first intent classifier with LLM fallback, routing to grounded QA, summarization, persona chat, and recommendation services.
- Built the evaluation/observability stack: **RAGAS**-based faithfulness/relevancy scoring, chunking A/B test harnesses, and self-hosted LLM tracing for token cost, latency, and quality.
- Served via async **FastAPI** with typed schemas and per-request latency/retrieval-score telemetry; containerized via Docker and systemd, with ownership of the GitHub repository and release process for independent deployment from `main`.

Junior AI/ML Engineer | Code Owls

July 2024 – November 2024

Chattogram, Bangladesh

- Developed and deployed **Generative AI workflows** using **LangChain** and **LangSmith**, enabling NLP-driven adaptive LLM response pipelines for production AI services.
- Researched and integrated emerging **AI technologies and NLP tools** — continuously exploring Generative AI trends to improve system intelligence and response quality.
- Built AI automation solutions using **Python, FastAPI, Git, and Linux (Ubuntu)** — supporting **data preprocessing**, model integration, and REST API deployment.
- Applied structured **A/B evaluation** and feedback loops in an **Agile** cross-functional team to improve AI model accuracy and user experience.

RESEARCH & PUBLICATION

A Resistome-Driven Ensemble Framework for AMR Phenotype Prediction | ECCT 2026

2026

Taylor & Francis Book Chapter ■ Co-author ■ Authors: Jamil As-ad, **Safaet Jaman Arman**, Kazi Injamamul Hoque, Md. Khaliluzzaman

- Designed the **Resistome Burden Index (RBI)** — a novel normalized cumulative feature quantifying aggregate resistance determinant load, capturing co-occurring mechanisms that binary gene features miss.
- Built the **R-Blend** heterogeneous **SVM-XGBoost Soft-Voting Ensemble** (margin-based + boosted-tree); evaluated on **12 antibiotic-pathogen datasets** from NCBI Pathogen Detection.
- Achieved **Avg. F1: 0.9421 | Recall: 0.9566 | ROC-AUC: 0.9656** — improved F1 in 8/12 datasets vs. prior methods; maximum gain **+13.8%** (*K. pneumoniae* meropenem).
- Ablation study confirmed RBI as the most critical component: removing it caused the largest recall drop (−0.0326).
- **Stack:** Python, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, Matplotlib, Google Colab. [GitHub]

EDUCATION

BSc in Computer Science & Engineering | International Islamic University Chittagong
Chattogram, Bangladesh

2019 – 2026 (Expected)

PROJECTS

ResistomeX — AMR Prediction Web App productizing the ECCT 2026 research

[\[GitHub\]](#) [\[Live Demo\]](#)

Python · Streamlit · Scikit-learn · XGBoost

- Productized the R-Blend ensemble (Decision Tree + Logistic Regression + XGBoost, soft voting, RBI feature) into a live tool; avg. **F1 of 0.9421** across 12 antibiotic-pathogen datasets.

AI Lesson Planner — LLM-powered lesson plan generator for educators

[\[GitHub\]](#)

Python · LangChain · FastAPI · Git

- Built a Generative AI system that automatically generates structured lesson plans — leveraging LLM pipelines for educational content automation.

House Price Prediction — End-to-end ML model — live deployed on production server

[\[GitHub\]](#) [\[Live Demo\]](#)

Python · Scikit-learn · Streamlit · Git

- Developed a regression-based ML model for house price prediction with full data preprocessing pipeline deployed live via Streamlit.

DevSearch — Developer Portfolio & Skill Matching Platform — Full-stack web app

[\[GitHub\]](#)

Django · HTML/CSS/JS · AWS · Git

- Built and deployed a developer skill matching and portfolio management platform on AWS.

TECHNICAL SKILLS

AI / ML	PyTorch · TensorFlow · Scikit-learn · XGBoost · LightGBM · CatBoost · NumPy · Pandas
LLM / RAG	LangChain · LangGraph · LangSmith · HuggingFace Transformers · PEFT/LoRA · FAISS · RAGAS · OpenAI API · Prompt Engineering
Languages	Python · SQL · C/C++
Frameworks	FastAPI · Django · Streamlit
Databases	PostgreSQL · MySQL · SQLite · ChromaDB
DevOps/Cloud	Docker · systemd · AWS · GitHub Actions · Git · Linux (Ubuntu)
Tools	Weights & Biases · MLflow · vLLM · Google Colab · Kaggle · Postman · NCBI AMRFinderPlus

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- 4th Place** — IIUC Intra-University Programming Contest, Autumn 2022.
- Kaggle** — Titanic: Machine Learning from Disaster; implemented full ML classification pipeline, achieved **Rank 1003** globally.
- Active competitive programmer on **Stopstalk** and **LeetCode**.

PUBLICATION REFERENCE

As-ad, J., Arman, S. J., Hoque, K. I., & Khaliluzzaman, Md. (2026). *A Resistome-Driven Ensemble Framework for Antimicrobial Resistance Phenotype Prediction*. In *Proceedings of ECCT 2026*. Taylor & Francis.